AI SYNERGY HACKATHON 2026 · ROUND 1 · CASE INVESTIGATION

LoopBack — the follow-up blind spot

Every “come back” said in the OPD becomes a dated, consented follow-up with a named owner — captured at the desk while the patient is still there.

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Document Problem–Solution Brief

01 Selected theme and case

Track 03, Healthcare: **“The Follow-Up Blind Spot.”** A district hospital runs high-volume outpatient clinics — about **900 OPD visits a day**. Clinicians can tell who needs to come back, but the hospital struggles to make sure those patients return at the right time with the right information. It wants fewer avoidable losses to follow-up, with **no unsafe automated medical decisions**.

28% of patients marked for follow-up did not return within the recommended window (3-month sample)

19% of records share one household phone — a plain reminder is unsafe

7% of records have an unreachable number — found only after the miss

~900 OPD visits a day; any per-patient step must take seconds

02 Stakeholders

Who benefits (primary)

- **Patients and caregivers** — the people who miss care when a “come back” is lost.

Who uses it daily

- **OPD desk nurse** — confirms the follow-up while the patient is present.
- **Records staff** — work the recovery list and make the calls.

Also involved

- **Clinicians** — write the instruction, set importance, settle flagged cases.
- **ASHA community health workers** — home follow-up where the patient agreed.
- **Hospital administration** — read the outcome and fairness figures.

Choosing **patients and caregivers** as the primary stakeholder is what moves the fix upstream — to the desk, while the patient is still present — rather than to a call centre after the miss.

03 The problem identified

When an OPD clinician asks a patient to come back, the request lives only as free text on a slip. Three linked failures follow:

Failure	What goes wrong	Evidence
The request is never captured	“R/V 2/52 with CBC”, “SOS”, “2 hafte baad” never become a date anyone can track. No system holds a due date, so nobody can see a miss.	CASE free-text, varies by clinician
Contact is checked too late	Shared and unreachable numbers are discovered only *after* the miss, when the patient has already gone home.	CASE 19% shared · 7% dead
Misses have no owner or order	A small records team cannot tell which missed visit to chase first, or how. Nothing says whom to call first.	CASE 28% miss the window

> **Problem statement.** An OPD “come back” exists only as free text. The hospital has no record of **who is due, by when, why it matters or how to reach them** — so the 28% who miss their window go unnoticed, and a small records team cannot tell which missed visits to chase first, or how.

04 Alternative problem statements considered

We tested four framings against the case before choosing one. Each was judged on whether it fixes the root cause, stays inside the case’s safety boundary, and is testable with data we already have.

Framing	Primary stakeholder	Verdict
Send automatic reminders	Patients	**Partial.** Needs a due date that does not exist yet; fails the quarter of patients on shared or dead numbers; does nothing for misses already piling up. **Kept as one component.**
Predict who will miss, or score clinical urgency	Administration	**Rejected.** No recorded outcomes to learn from; staff could not see why a score was given; judging urgency belongs to clinicians under the case boundary.
Solve transport and distance	Patients far away	**Out of V1.** Real (22% cited distance) but outside the OPD and dependent on funding. The worklist records the barrier so the case can be made later.
Record the follow-up properly, then track & recover the misses	Patients & caregivers; nurse & records staff	**Chosen.** Fixes the root cause, uses the one moment the patient is present to check contact and consent, and gives staff an order they can explain. Needs no clinical judgement from software. Buildable and testable in a week.

If a different stakeholder were primary: with the **records team** first, the answer drifts to a better call list — still blind to who is due. With **administration** first, to dashboards and prediction — with no outcomes to learn from. Putting **patients and caregivers** first is what moves the fix upstream.

05 Evidence and assumptions

CASE from the case file EXTERNAL independent source, cited ASSUMPTION believed, not yet verified
UNKNOWN could change the decision

Label	Claim	What it means for the design
CASE	28% of follow-up patients did not return within the window (3-month sample).	Baseline for the main measure: 72% on time.
CASE	Instructions are free text and vary by clinician.	No system holds a due date — the root of the problem.
CASE	19% share a household phone; 7% are unreachable.	Contact must be checked while the patient is present.
CASE	A small records team cannot prioritise every miss.	One ordered list, with a reason on every row.
EXTERNAL	A Cochrane review found text reminders increase attendance vs none — moving it ~11 points (67.8%→78.6%), similar to phone calls. [1]	Reminders are worth sending, but only one part of the answer.
EXTERNAL	Top reasons for missing: felt well 25%, distance 22%, no instructions/reminders 15%. [2]	Missing reminders is a real top-three cause; distance stays out of V1; every closed case records *why*.
EXTERNAL	DPDP Act 2023 s.6: consent must be free, specific, informed and unambiguous. [3]	Separate consent per contact route; nothing pre-ticked; date & role logged.
EXTERNAL	On the Gemini free tier, inputs may be used to improve products and read by reviewers. [4]	No patient identifiers leave the desk; simulated data only.
ASSUMPTION	20–30% of visits need follow-up → ~50–75 new misses a day at 28%.	~1,500–2,300 misses a month — too many to call without a ranked list.
ASSUMPTION	A clerk manages 40–50 contact attempts a day; the desk can spend 60–90 s per patient.	Order matters more than speed; confirmation must be one page with answers pre-proposed.
UNKNOWN	Which misses are clinically important.	Not guessed — the clinician ticks routine / important / must-not-miss.
UNKNOWN	Which contact routes patients accept; how varied real notes are.	Asked one by one at the desk; reader accuracy measured on de-identified notes in Round 2.

06 Why we chose this problem

More important

Every later step — reminders, recovery, measurement — depends on the record that is missing today. Fix the record and the rest becomes possible.

More tractable

Follow-up language is narrow: intervals, dates and a handful of shorthands (2/52, 6/12) that rules can read. No open-ended clinical understanding is required.

More testable

“Missed” and “returned” are already facts in registration data. We need no new outcome labels to measure success.

Within the safety boundary

The chosen framing needs **no clinical judgement from software** — importance comes from the clinician, and nothing is diagnosed or triaged. It respects every hard limit in the case file.

07 Proposed solution — LoopBack

One record, four steps. **A person confirms every decision**; the system only proposes, flags and keeps order.

Step	Owner	What happens
1 · Capture	Desk nurse	A rule reader turns the clinician’s words into a **proposed date** and a bring-list. It reads clinical shorthand, Hinglish and Hindi, ignores medicine courses (“x 14 days”), moves dates to days the clinic runs, and **flags anything it cannot read safely**. Nothing saves until the nurse confirms.
2 · Check contact	Desk nurse	While the patient is present: whose phone, is it shared, which language, and **consent for each route** (text, call, caregiver, ASHA visit). The patient leaves with a bilingual slip carrying the clinician’s return precautions word for word.
3 · Track	LoopBack	Each follow-up gets a window from the clinician’s importance level. A reminder goes two days before, with consent. Registration records the return; date arithmetic marks the rest **due / missed / returned** — never prediction.
4 · Recover	Records / ASHA	Misses enter **one ordered list** (importance, then days overdue), with a reason on every row. Contact uses only the agreed route; clinical questions go to a nurse; nothing closes without a reason code.

What the system does

Proposes dates, flags problems, does date arithmetic, orders the list, fills fixed message wording.

What people do

Confirm every follow-up, set importance, make the calls, answer clinical questions, close cases.

> **Optional second reader.** Gemini (free tier) can cross-check the rule reader on **de-identified text only**. Its answer is kept only where its evidence appears word for word in the instruction. Agreement is noted; disagreement becomes a decision for the nurse. **The system never** diagnoses, triages, changes treatment, gives medical advice, contacts anyone without consent, closes a case on its own, or sends identifiers to an AI service.

References — [1] Gurol-Urganci I, et al. Mobile phone messaging reminders for attendance at healthcare appointments. *Cochrane Database Syst Rev* 2013;(12):CD007458. [2] Das B, et al. Non-adherence to medical appointments among hypertension patients, public facilities, Punjab. *J Clin Hypertens* 2021;23(4):713–719. [3] Digital Personal Data Protection Act, 2023, s.6; DPDP Rules 2025. [4] Google, Gemini API Additional Terms of Service. All patients and outcomes shown are simulated; case figures are competition material.